

Sarwad Hasan Siddiqui

CS UNDERGRADUATE RESEARCHER · GPU & AI SYSTEMS · SOFTWARE ENGINEER

sarwad015@gmail.com · 01869965299 · github.com/Rockstatata
linkedin.com/in/sarwad-hasan-siddiqui · Google Scholar · Amar Ekushey Hall, KUET, Khulna

PROFILE

CSE undergraduate researcher at KUET building intelligent, high-performance systems at the intersection of AI, GPU computing and software engineering. Focus: AI systems, CUDA/HPC, efficient transformers, computer vision and HRI, with reproducible, measurable experiments. Aiming for graduate research in AI systems.

RESEARCH & PUBLICATIONS

- **Medora: A Consent-Aware, AI-Native Bilingual Platform for Healthcare Management and Consultation.** S. H. Siddiqui, A. Tahsin, K. S. Alam, S. I. Hossain. PUBLISHED
- Paper at *IEEE RO-MAN 2026* (human–robot interaction). PUBLISHED
- Poster paper at *IHCI 2026*. POSTER
- Submission to *IEEE ICRA 2027*. UNDER REVIEW
- Submission to *ICCIT 2027*. UNDER REVIEW
- **FLAG 2027 (ICASSP Grand Challenge)**: cross-lingual face–voice verification; verified FOP reproduction reaching 36.32% dev EER vs. 36.92% baseline. ONGOING
- **F2HF journal extension**: FFN-weighted head filtering for Vision Transformer explanations. **DriftRisk-LoRA**: drift-aware financial risk modelling (thesis). ONGOING

SELECTED PROJECTS

Medora
Next.js · FastAPI · PostgreSQL · LLMs
github.com/CSE-3200-System-Project/Medora
Bilingual, consent-governed health platform: policy chokepoint around assistive AI, review-gated prescription OCR, idempotent booking. 100+ commits.

Agun Panjika
Python · React · D3 · Three.js
github.com/Rockstatata/Agun-Panjika
NASA Space Apps 2026: harmonised MODIS/VIIRS fire history for Bangladesh with seasonal alerts and a two-week outlook.

Jagoo Bahee
TypeScript · Rust · Reticulum
github.com/tasmirz/jagoo-bahee
Censorship-resistant federated forum that survives a national blackout via signed envelopes over HTTP, mesh and LoRa. July Hackathon 2026 runners-up.

Dhara
PyTorch · BGE-m3 · LoRA
github.com/Rockstatata/Dhara
Bangla legal retrieval across 1,227 Acts; fine-tuned BGE-m3 + LoRA lifts Recall@10 from 0.067 (BM25) to 0.50.

Voyager-2
C++20 · OpenGL · GLSL
github.com/Rockstatata/Voyager-2
GPU solar-system renderer driven by NASA/JPL Horizons data: floating origin, shadow maps, LOD and a ray-traced view.

Interlink
React Native · TypeScript
github.com/Logarithm-Studio/Interlink-backend
AI agent automating daily and professional workflows; core engineer on app and backend at Logarithm Studio.

EDUCATION

B.Sc. in Computer Science & Engineering
Khulna University of Engineering & Technology (KUET)
2023 – 2027 · CGPA: 3.77 / 4.00 · Dean's List ×2

Higher Secondary Certificate (HSC)
Notre Dame College, Dhaka
2019 – 2021 · GPA: 5.00 / 5.00

Secondary School Certificate (SSC)
Adamjee Cantonment Public School, Dhaka
2015 – 2019 · GPA: 5.00 / 5.00

TECHNICAL SKILLS

Python · C/C++ · CUDA · PyTorch · Transformers / LoRA · Computer Vision · OpenGL / GLSL · TypeScript · React / Next.js · React Native · FastAPI / Django · PostgreSQL · Docker · Flex & Bison · Java / Swift

ACHIEVEMENTS

- ★ Runners-up — July Hackathon 2026
- ★ 3rd Place — Nascenia AI Hackathon (Kaggle)
- ★ 1st Runner-Up — Intl. Bridging Engineering & CS Competition 2025 (ILM AI)
- ★ Dean's List ×2 & Dean's Award, KUET
- ★ 3rd Runner-Up — Case Crack 2.0 (KBEC)
- ★ 2nd Place — Spark Fusion 2K21 (Hardware Design)

LANGUAGES

Bengali	Native
English	Professional

EXPERIENCE & LEADERSHIP

Undergraduate Researcher — 2025 – Present
KUET
HRI, healthcare AI, computer vision and multimodal research: Medora, RO-MAN, IHCI, ICRA/ICCIT submissions, FLAG 2027.

Junior Software Developer (Part-time) — Oct 2025 – 2026
AlgoSoft Technologies Limited
Mobile apps and Python services incl. the AI-native Daily Gratitude journal. Intern from Apr 2025.

Asst. Joint Secretary & Group Co-Organizer — 2026
TEDxKUET 2026
PR team: guest & speaker relations, logistics, promo videos reaching 50.5K+ views.

Software Engineer — Mar 2026 – Present
Logarithm Studio
Core engineer on Interlink (AI automation agent): React Native app and TypeScript backend.

Senior Executive — Sep 2025 – Present
Team Durbar · KUET Mars Rover
Member of KUET's rover team building rovers for international rover challenges.

Secretary of PR — Jun 2026 – Present
KUET Business & Entrepreneurship Club (KBEC)
Operational planning for events and workshops; logistics, communications and onboarding.